



# A wavelet neural operator based elastography for localization and quantification of tumors

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## ABSTRACT

**Background and Objectives:** The application of intelligent imaging techniques and deep learning in the field of computer-aided diagnosis and medical imaging have improved and accelerated the early diagnosis of many diseases. Elastography is an imaging modality where an inverse problem is solved to extract the elastic properties of tissues and subsequently mapped to anatomical images for diagnostic purposes. In the present work, we propose a wavelet neural operator-based approach for correctly learning the non-linear mapping of elastic properties directly from measured displacement field data.

**Methods:** The proposed framework learns the underlying operator behind the elastic mapping and thus can map any displacement data from a family to the elastic properties. The displacement fields are first uplifted to a high-dimensional space using a fully connected neural network. On the lifted data, certain iterations are performed using wavelet neural blocks. In each wavelet neural block, the lifted data are decomposed into low, and high-frequency components using wavelet decomposition. To learn the most relevant patterns and structural information from the input, the neural network kernels are directly convoluted with the outputs of the wavelet decomposition. Thereafter the elasticity field is reconstructed from the outputs from convolution. The mapping between the displacement and the elasticity using wavelets is unique and remains stable during training.

**Results:** The proposed framework is tested on several artificially fabricated numerical examples, including a benign-cum-malignant tumor prediction problem. The trained model was also tested on real Ultrasound-based elastography data to demonstrate the applicability of the proposed scheme in clinical usage. The proposed framework reproduces the highly accurate elasticity field directly from the displacement inputs.

**Conclusions:** The proposed framework circumvents different data pre-processing and intermediate steps utilized in traditional methods, hence providing an accurate elasticity map. The computationally efficient framework requires fewer epochs for training, which bodes well for its clinical usability for real-time predictions. The weights and biases from pre-trained models can also be employed for transfer learning, which reduces the effective training time with random initialization.

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## 1. Introduction

Solutions to the inverse problems are essential in various domains of science and engineering [1–3]. For example, in the field of biomedical engineering, identification of the elasticity of the healthy and pathological tissues from *in-vivo* deformation data involves solving an inverse problem [4]. Pathogens have been identified as one of the causes of alterations of healthy tissues to diseased ones. As the disease progresses with time, the pathogens

continuously change the stiffness of healthy tissues. Overall, these changes in stiffness are the collective response of various alternations at the molecular, cellular, and tissue levels. Thus in biomedical engineering, the determination of the stiffness distribution of soft tissues through inverse problems remains to be an active research field [5–7]. In the clinical setting, the primary approach is to detect any abnormality in the organ by hand palpation. Although hand palpation is a time-tested method, it is a qualitative method and heavily dependent on the skills of the clinician. Conversely, the rise of elastography as an alternative to hand palpation provides a quantitative elasticity map [8]. Due to advancements in imaging techniques, it is now possible to capture the deformation of the region of interest under loading, whether static or dynamic. Imaging techniques such as ultrasound, magnetic resonance imag-

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ing, or digital image correlation can provide the displacement or strain field in two or three dimensions [9–11]. Using these images, elastography aims to provide diagnostic information about diseases and aid clinicians in decision-making intraoperatively. Furthermore, the material properties acquired via elastography can also be used in the patient-specific virtual models of organs for use in surgical planning and training purposes [12].

The displacement or strain-based inverse problems have gained significant attention in the last decade and have been investigated thoroughly for medical imaging applications [1,13,14]. The current state-of-the-art consists of encoding the motion information using phase-sensitive imaging modalities and extracting the displacement field using cross-correlation, phase-tracking, mechanics-driven optimization based and recently employed deep-learning based methods [15–19]. The displacement field is further utilized to estimate the spatial distribution of the material parameters by solving an inverse problem of elasticity. The inverse problem is set up using the physical laws of mechanics, describing the system in the form of Partial Differential Equations (PDEs). For any system, the PDEs are formed using the equilibrium equations followed by the constitutive relations and boundary conditions. Direct and iterative methods have been utilized extensively in the past to solve the ill-posed inverse problem [16,20]. In direct approaches, the measured displacement field is directly substituted in the algebraic form of the governing PDEs. The PDEs are simplified using various assumptions related to material behavior, such as perfect incompressibility or local homogeneity of the material parameters. The direct approaches are quick in providing the results; however, they are prone to errors originating from the measured noisy data, material behavior assumptions, and large strain gradients [16]. The iterative methods, on the contrary, start with an initial guess of unknown material parameters and solve a PDE-constrained optimization problem [21,22]. In most cases, the Finite Element Method (FEM) has been used to find a unique solution to the PDEs that are parameterized by the latest estimate of material parameters in each iteration [16,20]. The model computed displacement field is compared with the measured one, and the error is minimized by updating the material parameters. Although FEM provides the liberty to incorporate complexity in terms of physics or geometry, these methods are computationally expensive; at each iteration, one has to solve a forward problem followed by sensitivity analysis. The computational cost associated with these sub-operations in iterative approaches increases multifold with an increase in the number of unknown parameters [21]. To handle the noisy displacement data in both approaches, regularization as prior knowledge is used [23]; however, the penalty factor associated with the regularization is often selected in an *ad hoc* manner. Overall, the real-time prediction of the elasticity using the aforementioned methods is extremely challenging.

In recent years, neural networks have gained significant interest in engineering applications for solving both forward as well as inverse problems [24–27]. Using a combination of the linear weighted sum of inputs with nonlinear activation functions, the neural networks can learn almost everything [28]. Deep learning-based Machine Learning (ML) techniques can extract useful features from the high dimensional input space and utilize the same to predict the output for a given task. Additionally, the computational bottleneck arising in the FEM analysis is skipped since the outputs are directly mapped to inputs without iterations. The deep learning-based approaches have demonstrated their applicability in various domains such as natural language processing [29], image classification [30], fracture mechanics [31], medical diagnosis [32–35], etc. However, data-driven deep learning models often fail to generalize to out-of-distribution data. Two possible alternatives that address this issue include physics-informed deep learning [25,36–38] and operator learning [39–41]. While the former

regularizes the training stage by incorporating known governing physics into the loss function, the latter attempts to directly learn the operator by using the universal approximation theorem for operators [28]. However, the mechanism of biological processes is often complex and remains elusive, due to which the exact governing physics of underlying biology is not known. In such cases, neural operators can learn the underlying governing physics and can provide reliable predictions. The focus of this study is on operator learning, and hence, the discussion here onwards is restricted to operator learning algorithms only. In particular, we focus on the recently proposed Wavelet Neural Operator (WNO) because of its superior accuracy as illustrated in [40].

In this work, we propose a novel approach that can estimate the elastic properties from measured displacement fields. To that end, the superior expressive capability of WNO is exploited. The WNO uses wavelet basis functions to localize the frequency information of the input over spatial coordinated, allowing a unique way to learn the inverse operator that maps the displacement field to the elastic field. Once the underlying operator is learned, the WNO can map any input displacement field from a family to its corresponding elastic properties. The proposed approach is optimization free and hence, applicable in real-time. Overall, the novel features of the proposed framework can be encapsulated in the following comments:

1. The proposed deep neural operator-based elastography directly learns the operator responsible for mapping the displacement fields to elastic fields. This is very convenient when the exact physics for the corresponding stiffness-displacement relationship is not known.
2. The proposed framework maps the input displacement field data to the elastic field by circumventing the steps of obtaining strain and stress fields from displacement data. This reduces the computational bottlenecks involved in the computation of gradients of noisy displacement fields.
3. The multi-resolution wavelet decomposition treatment of the input displacement field helps in separating the effect of observation noise from data and better prediction in actual scenarios.

We illustrate the efficacy of the proposed approach on synthetic and experimental data. The results obtained are encouraging and illustrate the possible application of the proposed approach in clinical settings. Further, a noise sensitivity analysis has also been performed to demonstrate the robustness of the framework.

The remainder of the paper is arranged as follows: in Section 2, the problem of the proposed work is defined. Then, the proposed framework for directly obtaining the mapping between the displacement and elastic fields using the WNO is illustrated. In Section 3, first, the data curation and the description of different elasticity distributions used for simulating the displacement field are provided. Then the results of the identification of the elasticity fields from displacement data are demonstrated. Experimental verification is also carried out to support the robustness of the proposed framework. Further, the sensitivity of the proposed framework for varying levels of noise is illustrated. Finally, in Section 4, the salient features of the proposed framework are revisited, and future directions on the extension of the proposed work are discussed.

## 2. Methods

### 2.1. Problem statement

The purpose of solving elastography-based inverse problems is to provide a non-invasive estimate of the elastic properties of soft materials. The broad application of elastography is to aid clinicians

in better pre-operative planning by providing quantitative information regarding the lesions. For instance, the lesion shape, size, elasticity, and degree of elasticity distribution are some of the crucial biomarkers that can be exploited for diagnostic and therapeutic purposes [42]. The majority of this information about the lesion is determined by solving an ill-posed inverse problem in a traditional way by utilizing displacement field data. Alternatively, the capabilities of WNO could be leveraged to directly provide a functional mapping between the useful features derived from the field data and quantitative metrics such as elasticity. To train and test the efficacy of the WNO for solving the inverse problems of elastography, a large set of displacement field data was generated. The data was generated by solving a Boundary Value Problem (BVP) of elasticity, which consisted of a stiff lesion embedded within a soft background. Mathematically, the BVP can be posed by considering a smoothly bounded region  $\Omega \in \mathbb{R}^d$  representing a solid domain bounded by  $\Gamma$  such that  $\Omega \cup \Gamma = \bar{\Omega}$  and  $\Omega \cap \Gamma = \phi$  in  $d$ -dimensions. The boundary consists of Dirichlet and Neumann boundaries such that  $\Gamma = \Gamma_D \cup \Gamma_N$  and  $\Gamma_D \cap \Gamma_N = \phi$ . The balance of linear momentum yields field equations for a linear elastic material:

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \rho \ddot{\mathbf{u}} \quad (1)$$

where  $\boldsymbol{\sigma}$  is the symmetric stress tensor,  $\mathbf{b}$  is the body force per unit volume,  $\rho$  is the mass density, and  $\ddot{\mathbf{u}}$  is the acceleration ( $\square$  operator denotes differentiation with respect to time). In the absence of body forces, Eq. (1) can be re-written for a quasi-static analysis (a.k.a. equilibrium equation) as:

$$\nabla \cdot \boldsymbol{\sigma} = 0 \quad (2)$$

The constitutive relation for an isotropic material can be given as:

$$\boldsymbol{\sigma} = 2\mu \boldsymbol{\varepsilon} + \lambda(\nabla \cdot \mathbf{u})\mathbf{I} \quad (3)$$

where,  $\lambda(x, y)$  and  $\mu(x, y)$  are the nonnegative Lamé parameters and  $\mathbf{I}$  is the identity tensor. The Lamé parameters can also be expressed in terms of Young's modulus ( $E$ ) and Poisson's ratio ( $\nu$ ) as:

$$\mu = \frac{E}{2(1+\nu)} \quad ; \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \quad (4)$$

Further, the strain displacement relation can be given as:

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T) \quad (5)$$

By utilizing Eq. (5) in Eq. (3) and finally substituting the resulting expression in Eq. (2), one can rewrite the equilibrium equation as:

$$\nabla \cdot (\mu(\nabla \mathbf{u} + \nabla \mathbf{u}^T) + (\lambda \nabla \cdot \mathbf{u})\mathbf{I}) = \mathbf{0} \quad (6)$$

The final set of field equations, constitutive relations along with Dirichlet (Eq. (7)c) and Neumann (Eq. (7)d) boundary conditions is given by [43]:

$$\begin{cases} \nabla \cdot [\mu(\nabla \mathbf{u} + \nabla \mathbf{u}^T) + (\lambda \nabla \cdot \mathbf{u})\mathbf{I}] = \mathbf{0} & \text{in } \Omega \\ \boldsymbol{\sigma} = \lambda \text{tr}(\boldsymbol{\varepsilon})\mathbf{I} + 2\mu \boldsymbol{\varepsilon} & \text{in } \Omega \\ \mathbf{u} = \mathbf{u}_0 & \text{on } \Gamma_D \\ \boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{h}_0 & \text{on } \Gamma_N \end{cases} \quad (7)$$

By expressing Lamé parameters in terms of Young's modulus ( $E$ ) and Poisson's ratio (if assumed constant), Eq. (7) represents the partial differential equations parameterized by the material parameter field,  $E(x, y)$ . In the majority of engineering applications, a forward problem is solved for an unknown displacement field ( $\mathbf{u}$ ) while the value of the material parameters is known *a priori*. For inverse problems, the displacement field is already known in the partial or complete domain, and Young's modulus or elasticity field becomes unknown. In this work, the functional mapping between

the displacement and elasticity fields has been learned using a neural operator rather than the traditional approaches utilized to solve the inverse problem. The WNO has been trained using the displacement field data generated by assuming different types of distributions for the elasticity field. The description of WNO is provided in the next section.

## 2.2. Displacement to elastic field mapping using wavelet neural operator

The WNO is a data-driven framework for learning the mapping between two infinite-dimensional function spaces. Unlike normal neural networks (fully connected networks and convolutional networks), which can only learn a single mapping between given input and output, the WNO learns the underlying operator (i.e., the family of input-output mappings). In elastography practices, the functional input-output can be viewed as input displacement fields and output stiffness field. Once the learning process is over, a single WNO can perform predictions for different functional inputs without retraining the network. This is very important in elastography applications, where the same pre-trained network is expected to provide reliable estimates of the elastic properties of the tissues for different displacement fields (e.g., various parts of organs and different tissue conditions across patients).

To present mathematically, we reconsider the smooth  $d$ -dimensional domain  $\Omega$  and take a spatial coordinate  $x \in \Omega$ . Over the spatial coordinate, we define the displacement field  $u(x)$  and the elastic field  $E(x)$  such that  $E(x) = \mathcal{D}(u(x))$ . Since there will exist multiple displacement and elastic fields (which are itself a function), we define two function spaces such that  $u(x) \in \mathcal{U}$  and  $E(x) \in \mathcal{E}$  [44]. Our aim is to learn the nonlinear operator  $\mathcal{D} : u(x) \mapsto E(x)$ , which maps the input displacement field  $u(x)$  to elastic field  $E(x)$ . We denote the dimensions displacement field  $u(x)$  and elastic fields  $E(x)$  as  $d_u$  and  $d_e$ , respectively. In our case, we want to utilize displacement fields in the  $x$  and  $y$  direction only, and we want to predict a single elastic field. Therefore, in our case  $d_u = 4$  and  $d_e = 1$ . However, to preserve the generality, we continue to denote the dimensions of  $u(x)$  and  $E(x)$  as  $d_u$  and  $d_e$ , respectively. Also, we want to supply the information on grids to the network. Thus we concatenate the grids with input and denote  $a(x) = \{u(x), x\}$ . For training the network, we consider that we have access to  $N$  pairs (number of training samples) of the displacement-elastic fields  $\{a_j, E_j\}_{j=1}^N$ . We will approximate true operator  $\mathcal{D}$  by a neural network as,

$$\mathcal{D}_{\theta_{NN}} : \mathcal{U} \mapsto \mathcal{E}, \quad (8)$$

where  $\theta_{NN}$  denotes the weights of the neural network. In general, the grid size  $x \in \Omega$  will have certain finite dimensions (e.g., the resolution of medical images). Thus in our formulation, we define a finite-dimensional solution space by  $n$  point discretization  $\{x_p\}_{p=1}^n$  of the domain  $\Omega$ . To be able to learn the intrinsic features of the input displacement fields, we first lift the inputs  $u(x)$  to some high dimensional space  $d_v$  using a local transformation  $P : a(x) \mapsto v_0(x)$ . This high-dimensional transformation is equivalent to increasing the number of channels in simple convolution neural networks (CNN). Although the local transformation  $P$  can be modeled as a shallow or dense fully connected neural network (FNN) or as a CNN, we use a  $1 \times 1$  convolution. On the lifted space,  $l$ -number of iterations using the transformation  $v_{j+1} = G(v_j)$  is carried out, where the outputs of each iteration have the dimension  $d_v$ . These iterations resemble the usual concepts of hidden layers in neural networks; however, the hidden layers are defined to follow the operator theory in functional analysis [44]. At the end of the  $l$ -iterations, a second local transformation  $Q : v_l(x) \mapsto E(x)$  is applied to obtain the final  $d_e$ -dimensional elastic field  $E(x)$ . The step-wise

updates  $G(v_j)$  mentioned above are defined as follows,

$$G(v_j)(x) := \varphi\left((K(a; \phi) * v_j)(x) + Wv_j(x)\right); \quad x \in \Omega, \quad j \in [1, l], \quad (9)$$

where  $\varphi(\cdot)$  is a scalar-valued non-linear activation function,  $\phi \in \theta_{NN}$  is the kernel parameters,  $W: \mathbb{R}^{d_v} \rightarrow \mathbb{R}^{d_v}$  is a linear transformation, and  $K$  is the integral operator. Here the integral operator  $K$  is defined as,

$$(K(a; \phi) * v_j)(x) := \int_{\Omega} k(a(x), x, \xi; \phi) v_j(\xi) d\xi; \quad x \in \Omega, \quad j \in [1, l]. \quad (10)$$

The aim is to learn the kernel  $k(a(x), x, \xi; \phi)$  so that by performing the kernel integration in Eq. (10) the required elastic field can be obtained. In the proposed WNO-based elastography, the kernel  $k(\cdot)$  is learned in the wavelet space, i.e., the integral Eq. (10) is performed on the wavelet decomposed coefficients rather than the direct input fields  $a(x)$ . To create the wavelet space from the input, the wavelet decomposition is performed on the lifted input  $v_j(x)$ . If  $\psi(x)$  denotes the orthonormal mother wavelet, then the forward and inverse wavelet transforms  $\mathcal{W}(\cdot)$  and  $\mathcal{W}^{-1}(\cdot)$ , respectively, of the lifted variables  $v_j(x)$  are defined as follows [45],

$$\begin{aligned} (\mathcal{W}v_j)(\alpha, \beta) &= \int_{\Omega} \Gamma(x) \frac{1}{|\alpha|^{1/2}} \psi\left(\frac{x-\beta}{\alpha}\right) dx, \\ (\mathcal{W}^{-1}(v_j)_w)(x) &= \frac{1}{C_{\psi}} \int_0^{\infty} \int_D (v_j)_w(\alpha, \beta) \frac{1}{|\alpha|^{1/2}} \tilde{\psi}\left(\frac{x-\beta}{\alpha}\right) d\beta \frac{d\alpha}{\alpha^2}, \end{aligned} \quad (11)$$

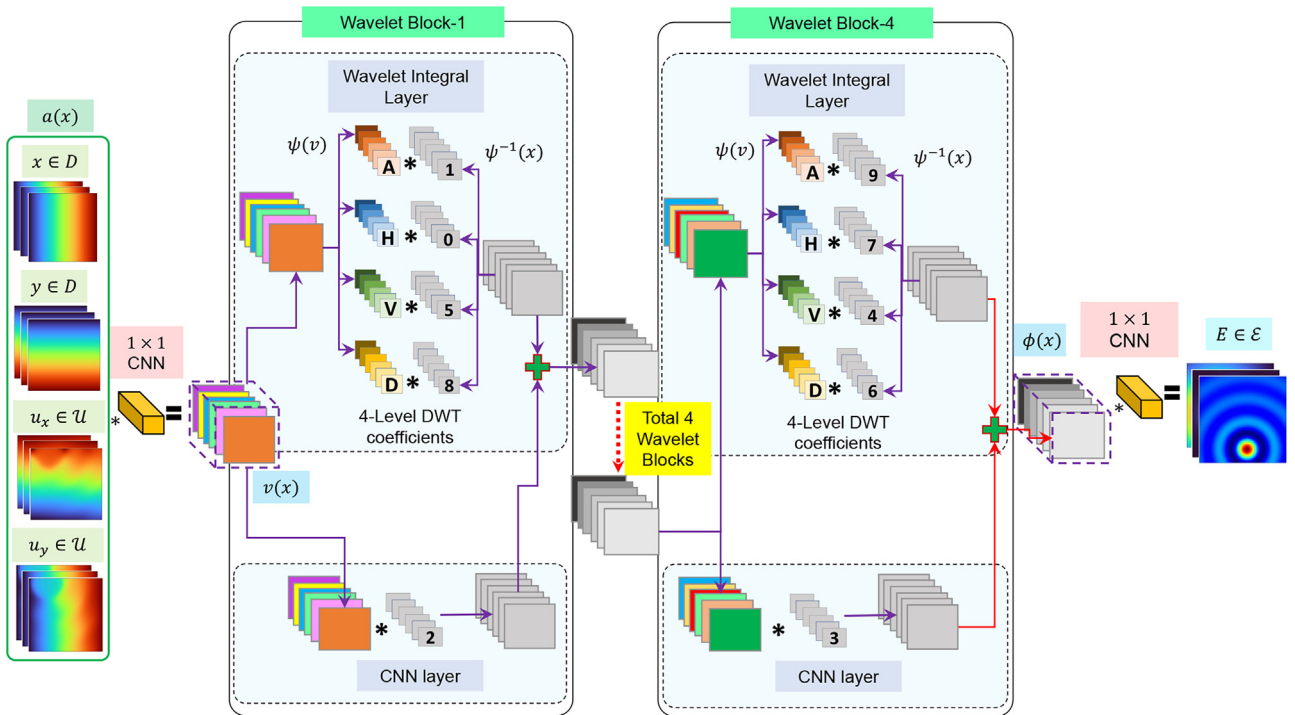
where  $\alpha$  and  $\beta$  are scaling and translational parameters of wavelet decomposition,  $(v_j)_w$  is the wavelet decomposed coefficients, and  $\tilde{\psi}(\cdot)$  is the scaled and translated version of the mother wavelet. The term  $0 < C_{\psi} < \infty$  is called the admissible constant. More details on the wavelet decomposition can be found in [45]. Using the convolution theorem, the integral in Eq. (10) over the wavelet domain is expressed as

$$(K(a; \phi) * v_j)(x) = \mathcal{W}^{-1}(\mathcal{W}(k_{\phi}) \cdot \mathcal{W}(v_j))(x); \quad x \in \Omega. \quad (12)$$

Here,  $\mathcal{W}(k_{\phi})$  represents the wavelet decomposition of the neural network kernel  $k_{\phi}$ . To simplify the implementation, it is advantageous to define the kernel  $k_{\phi}$  directly in the wavelet space, i.e.,  $R_{\phi} = \mathcal{W}(k_{\phi})$ . Thus, the wavelet kernel integration in Eq. (12) is rephrased as,

$$(K(\phi) * v_j)(x) = \mathcal{W}^{-1}(R_{\phi} \cdot \mathcal{W}(v_j))(x); \quad x \in \Omega. \quad (13)$$

The wavelet decomposition in Eq. (11) is usually expensive since the scale and translation parameters  $\alpha$  and  $\beta$  can be infinite dimensional. In order to reduce the computational and memory redundancy, we construct the wavelet space using the discrete wavelet transform (DWT) algorithm. In the DWT algorithm, the mother wavelet is generally modified to calculate the wavelet coefficients at scales with powers of two [45]. In DWT, we denote the scale and translational parameters by  $\ell$  and  $\tau$ , respectively. Then by fixing the parameter  $\ell$  to some integer and shifting the  $\tau$ , one can obtain the DWT of the inputs  $v_j(x)$ . Since our inputs are essentially images (second-order tensors), the DWT provides four wavelet coefficients, which are the approximate, horizontal, vertical, and diagonal components of the images. At each level, the dimension of the input is halved due to the conjugate symmetry. Once the DWT with  $\ell$ -level of decomposition is done, the length



**Fig. 1. Architecture of the proposed operator-elastography framework.** The discretized domain  $x, y$ , and the corresponding displacement fields  $u_x$  and  $u_y$  are concatenated to create a 4-channel input  $a(x)$ . The 4-channel input is lifted to a  $d_v$ -channel space using a  $1 \times 1$  CNN. The lifted inputs  $v(x)$  are passed through 4-wavelet blocks, which yields the output  $\phi(x)$  with  $d_v$ -channel. The  $d_v$ -channel output  $\phi(x)$  is transformed to the final 1-channel elasticity field  $E(x)$  using another  $1 \times 1$  CNN. Each wavelet block consists of a wavelet kernel integral layer and a convolutional neural network (CNN) layer with  $1 \times 1$  kernel. The wavelet integral layer learns the kernel  $k(a(x), x, \xi; \phi)$ , and the CNN layer performs the linear transformation  $Wv_j(x)$ . The wavelet integral layers first perform the multilevel wavelet decomposition of the input  $v(x)$  to obtain the approximate (A), horizontal (H), vertical (V), and diagonal (D) coefficients of inputs at different decomposition levels. Then the weight tensors  $R_{\phi}$  (one for each coefficient) are multiplied with the decomposed coefficients of the last decomposition level. After the multiplication, the inverse wavelet transform is performed on the convolved coefficients to produce the output  $\phi(x)$ . Suitable activation functions are applied to the output.

of the support width of wavelet coefficients can be calculated as  $d_l = n_s/2^\ell + (2n_w - 2)$ , where,  $n_s$  is the input size,  $\ell$  is the decomposition level, and  $n_w$  is the order of the vanishing moment of the mother wavelet (e.g., for Daubechies-6 wavelet, the vanishing moment is 6).

The parameterization of the kernel  $k_\phi(\cdot)$  is done at the highest scale of the DWT. Thus, if  $d_l$  denotes the dimension of the wavelet coefficients at the last level of DWT, then for the input  $v_j(x)$ , the decomposed output  $\mathcal{W}(v_j; \ell)$  will have the dimension  $d_l \times d_v$ . To be able to perform the integration in Eq. (13), we define a weight tensor  $R_\phi(\ell)$  with the size  $d_l \times d_v \times d_v$ . Finally the kernel multiplication  $(R_\phi \cdot \mathcal{W}(v_j))(x)$  is expressed as follows,

$$(R \cdot \mathcal{W}(v_j; \ell))_{t_1, t_2} = \sum_{t_3=1}^{d_v} R_{t_1, t_2, t_3} \mathcal{W}(v_j; \ell)_{t_1, t_3}; \quad I_1 \in [1, d_l], \quad I_2, I_3 \in d_v. \quad (14)$$

In the above equation, the summation is performed between the last dimension of  $\mathcal{W}(v_j; \ell)$  and the second dimension of the  $R_\phi(\ell)$ . Since the DWT provides us with four decomposed coefficients, we need four independent weight tensors  $R_\phi^i(I)$ ,  $i = 1, \dots, 4$ , to effectively learn the operator mapping between the displacement and elasticity fields. The overarching framework of the proposed WNO is presented in Fig. 1. The uplifting dimension is analogous to channels in a CNN and varies across problems. Usually higher the uplift dimension easier is to learn, but too high may lead to overfitting. The vanishing moments of a wavelet should be selected based on the input displacement field data. The vanishing moments represent the smoothness of a wavelet. Ideally, if the images have high spatial variations, then wavelets with low-order (in WNO,  $< 4$ ) vanishing moments should be utilized to capture the local features. Similarly, for a smooth image, a wavelet with higher ( $> 4$ ) vanishing should be utilized to capture the higher polynomial characteristics of an image. However, in this work, we observe that too high ( $> 14$ ) vanishing moments cause the model to overfit the test data. Also, in practical applications, we want the model to generalize over a large set of unseen data. Thus, a balance can be achieved by setting the vanishing moments between 4–10. The selection of wavelet decomposition level  $\ell$  depends on how fine the kernel size we want. For e.g., if the input has a spatial dimension  $64 \times 64$  and if we want a kernel of sizes 4 and 8, then we need a decomposition level of 4 and 3, respectively. The number of wavelet blocks depends on the complexity of the underlying operator to be learned. However, we observe the number of wavelet blocks usually saturates within 4–5 blocks.

### 3. Results

#### 3.1. Data curation

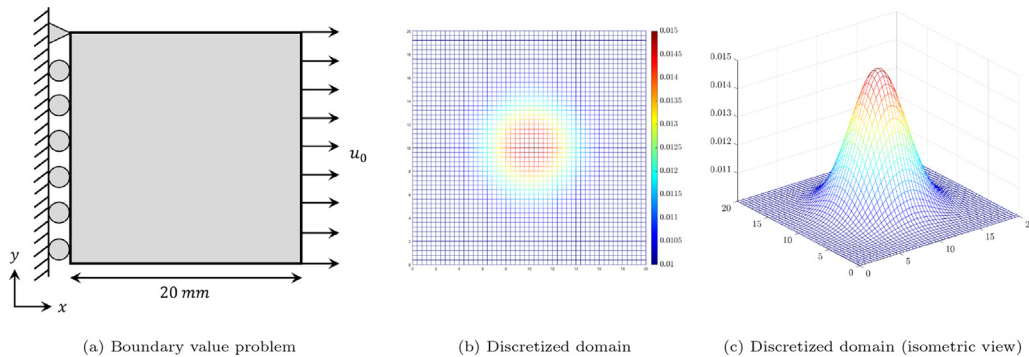
For the present case, a two-dimensional (2D) compressible forward problem of elasticity was solved using the FEM by assuming Young's modulus,  $E(x, y)$ , as a heterogeneous field [43]. The 2D boundary value problem, along with the boundary conditions, is given in Fig 2a. The governing differential equations, the constitutive relations, and boundary conditions for the same in the absence of body forces are stated in Eq. (7). A square domain measuring  $20 \text{ mm} \times 20 \text{ mm}$  is discretized with a resolution of  $51 \times 51$  so as to minimize the discretization errors. For all the simulations, a constant Poisson's ratio of 0.3 was assumed, while a heterogeneous distribution for Young's modulus was considered. The Young's modulus field was represented using the nodal values and interpolated using the same basis functions as used for the displacements (refer to Figs. 2b and 2c). A displacement boundary condition,  $u_0 = 0.2 \text{ mm}$  ( $\sim 1\%$  strain) in  $x$ -direction, was enforced on the right boundary while the left boundary was constrained with roller support. Additionally, the top point of the left boundary was constrained with hinge support in order to restrict the rigid body motion. The rest of the boundaries (top and bottom) were kept as traction free ( $\mathbf{h}_0 = 0$ ). The BVP was solved to generate the displacement field in  $x$ - and  $y$ -directions. An in-house parallelized MATLAB script was utilized to solve the BVP using FEM. The following section describes the different types of distributions assumed for Young's modulus. A total of 500 realizations were generated for each of the distribution types by sampling the different parameters governing their spatial distribution from a uniform distribution. For training and testing the WNO, a total of 500 pairs of labeled data consisting of spatial coordinates ( $\mathbf{x}$  and  $\mathbf{y}$ ), displacement field data ( $u_x$  and  $u_y$ ), and respective Young's modulus distributions were utilized.

#### 3.1.1. Inclusion

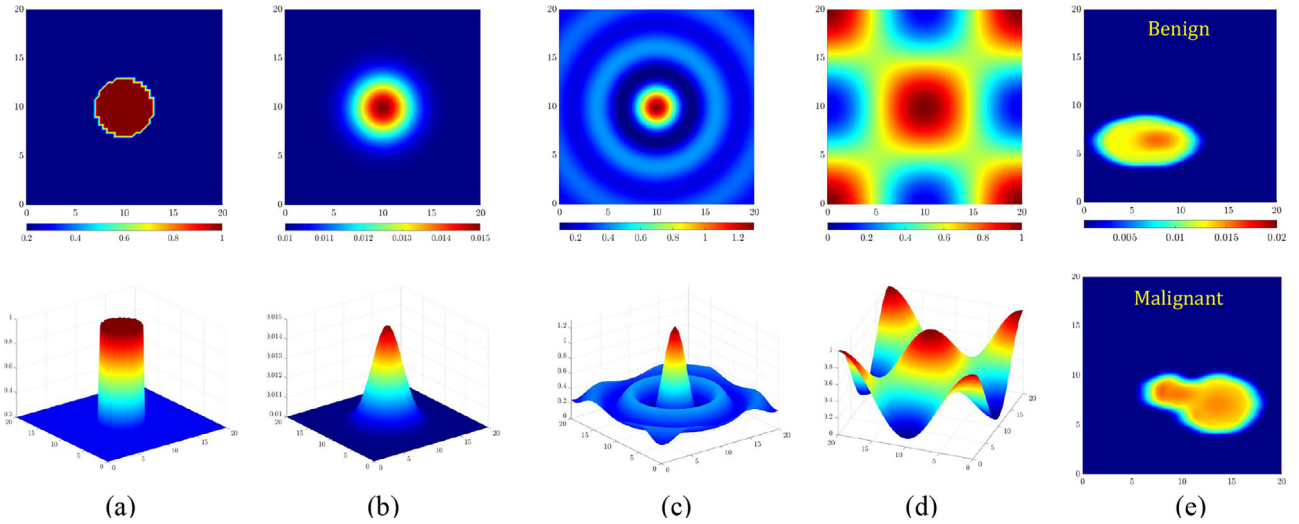
A distribution depicting a hard inclusion of fixed radius ( $r = 3 \text{ mm}$ ) embedded within the soft matrix was generated by sampling the center of inclusion from a uniform distribution as:

$$(x_c, y_c) \sim \mathcal{U}(0, l) \quad (15)$$

where  $l$  is the length of the square domain. One of the representative distributions (out of a total 500) thus generated is shown in Fig. 3a.



**Fig. 2.** (a) Definition of the boundary value problem and (b) discretized square domain along with nodal distribution of representative material property (in MPa) along with its (c) isometric view used for generating the displacement field data. (all axes are in mm).



**Fig. 3.** Representative top (top row) and isometric (bottom row) views of elasticity distributions (in MPa) for (a) hard inclusion, (b) Gaussian inclusion, (c) sinc distribution, and (d) 2D sinusoidal. (e) Tissue-mimicking phantoms embedded with benign (top) and malignant (bottom) tumors. (all axes are in mm).

### 3.1.2. Gaussian distribution

The following smooth variation for Young's modulus (kPa) in terms of a single Gaussian was assumed:

$$E(x, y) = 10 + 5 \exp \left[ -\left( \frac{x - x_c}{\sigma} \right)^2 - \left( \frac{y - y_c}{\sigma} \right)^2 \right] \quad (16)$$

where  $(x_c, y_c)$  and  $\sigma$  denote the center and standard deviation of the Gaussian, respectively. To generate the training data, the mean and standard deviation of the Gaussian was sampled from a uniform distribution:  $\{x_c, y_c\} \sim \mathcal{U}(0, l)$  and  $\sigma \sim \mathcal{U}(3, 10)$ . A total of 500 such realizations were generated for the elasticity distribution. One of the representative realizations for such elasticity distribution is shown in Fig. 3b.

### 3.1.3. Sinc distribution

Adding more complexity to the elasticity field, a normalized sinc function was utilized to generate the elasticity distribution (in MPa) as:

$$E(x, y) = 0.3 + \frac{\sin(\lambda \pi r)}{\lambda \pi r} \quad \forall \quad r \neq 0 \quad (17)$$

where  $r(x, y)$  is the distance of every point from the center  $(x_c, y_c)$  of the distribution and  $\lambda$  is a factor controlling the spatial frequency of the distribution. Both the parameters, the center and  $\lambda$ , were sampled from a uniform distribution:  $(x_c, y_c) \sim \mathcal{U}(0, l)$  and  $\lambda \sim \mathcal{U}(0.3, 0.6)$ . Additionally, a factor of 0.3 was added to the assumed distribution to avoid the negative values of elasticity. One of the generated realizations is shown in Fig. 3c.

### 3.1.4. 2D Sinusoidal distribution

The elasticity field of the domain was generated using a two-dimensional sinusoidal function given by:

$$E(x, y) = 0.1 + 0.45 \left( 1 + \cos \left( \frac{\lambda \pi x}{l} \right) \cos \left( \frac{\lambda \pi y}{l} \right) \right) \text{MPa} \quad (18)$$

where the factor  $\lambda$  controls the spatial frequency and was sampled from a uniform distribution,  $\lambda \sim \mathcal{U}(2, 4)$ . One of the generated realizations is given in Fig. 3d.

### 3.1.5. Benign and malignant tumor identification

Studies have shown that benign tumors have identifiable, regular boundaries and uniform elasticity distribution. In contrast, a

malignant tumor has irregular borders, abnormal growth, and heterogeneous elasticity distribution. In light of these findings, a recent study by Patel et al., [46] considered similar kind of distributions for the benign and malignant tumor classes. In the present work, similar distributions are assumed to represent benign and malignant tumors embedded in soft healthy tissue. The distributions (in MPa) were generated by superimposing two elliptic inclusions of different sizes given as:

$$E(x, y) = G \left( \sum_{i=1}^2 \chi_i \left[ E_0 + \hat{E}_i \left\{ 1 - \left( \frac{x - X_i}{a_i} \right)^2 \right\} \left\{ 1 - \left( \frac{y - Y_i}{b_i} \right)^2 \right\} \right] + 1 - [\chi_1 + \chi_2 - \chi_1 \chi_2] \right) \quad (19)$$

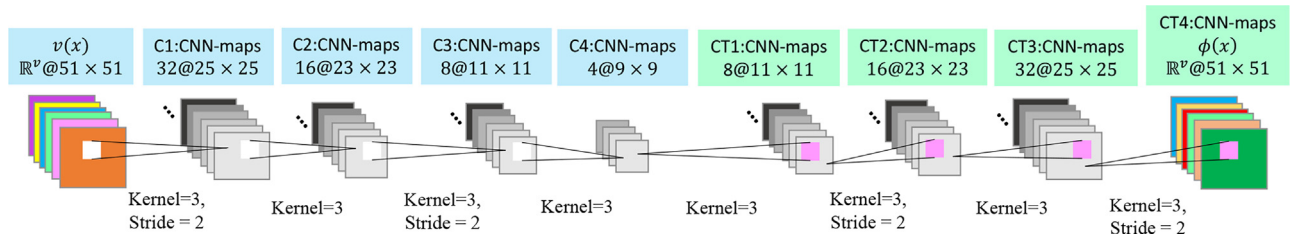
where  $\{(X_1, Y_1), (a_1, b_1)\}$  and  $\{(X_2, Y_2), (a_2, b_2)\}$  are the coordinates of centers and semi-major and semi-minor axes of the large and smaller ellipses, respectively,  $G$  is the 2D Gaussian filter with a standard deviation of 0.5, and  $\chi$  is an indicator function. The reader is referred to Ref[46], for a detailed description of the distribution. One of the generated distributions each for benign (top) and malignant (bottom) tumors is shown in Fig. 3e.

### 3.2. Learning of elasticity field of example problems through WNO

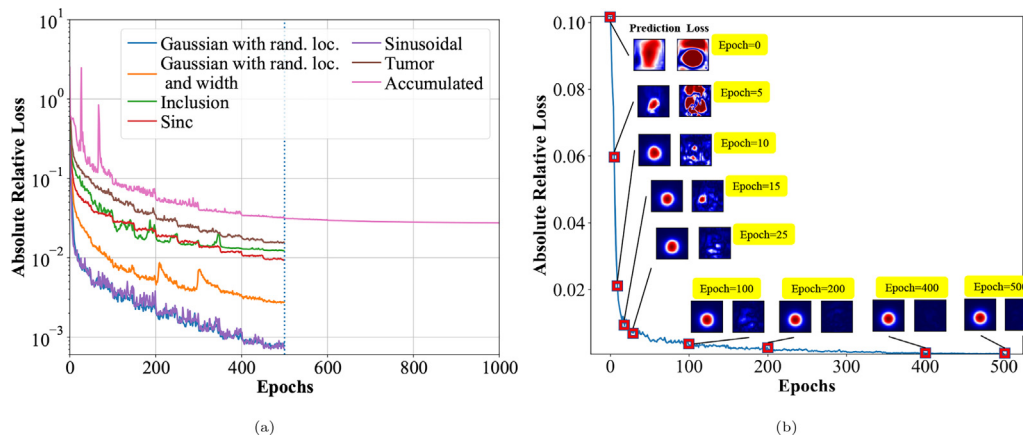
In this section, the results of the identification of the elasticity fields of undertaken examples are demonstrated. The training of the WNO architecture is done by using a total of 500 epochs for individual datasets and 1000 for the accumulated datasets. An initial learning rate of 0.001 with a step scheduler of step size 50 and a multiplication factor of 0.75 is utilized in both cases. In all examples, the *Daubechies* wavelet is employed to perform the decomposition, and the Gelu activation function is employed to learn the nonlinear operators. More details on the example-specific architectural parameters are provided in Table 1. For comparison with other data-driven deep learning-based approaches, a convolution neural network (CNN) based encoder-decoder similar to the Deep Learning Radiomics of elastography (DLRE) in [33] is constructed. The encoder and decoder consist of four CNN, and four transposed convolution layers, whose details are provided in Fig. 4. Further, we refer to this encoder-decoder network as one CNN-Block and use it to perform an ablation study by replacing the four Wavelet Blocks in Fig. 1 with four of these CNN-Blocks. The total number of epochs, learning rate, dataset size, and scheduler are kept the

**Table 1**  
Description of the WNO architecture utilized in present work.

| Examples             | Wavelet | Level | Layer dimensions |     |   |     | Data-Size |         |
|----------------------|---------|-------|------------------|-----|---|-----|-----------|---------|
|                      |         |       | P ( $d_r$ )      | Q   | W | WIL | Training  | Testing |
| Ex. 3.1.1            | db4     | 4     | 64               | 128 | 4 | 4   | 400       | 60      |
| Ex. 3.1.2            | db6     | 3     | 48               | 128 | 4 | 4   | 400       | 60      |
| Ex. 3.1.3            | db6     | 2     | 64               | 128 | 4 | 4   | 400       | 60      |
| Ex. 3.1.4            | db6     | 3     | 48               | 128 | 4 | 4   | 400       | 60      |
| Ex. 3.1.5            | db6     | 3     | 48               | 128 | 4 | 4   | 400       | 60      |
| Accumulated training | db8     | 3     | 64               | 128 | 4 | 4   | 1800      | 100     |



**Fig. 4. Schematic of the CNN encoder-decoder elastography.** The input  $v(x)$  denotes the lifted input  $a(x)$ , which is obtained using a  $1 \times 1$  convolution in a similar manner as the proposed operator elastography. Then four forward CNN and four transpose CNN layers are used to construct an encoder-decoder network. This provides the output  $\phi(x)$ , which is finally transformed to the final elasticity field using  $1 \times 1$  convolution, as in Fig. 1. Throughout the article, we denote the encoder-decoder network as a one CNN-Block.



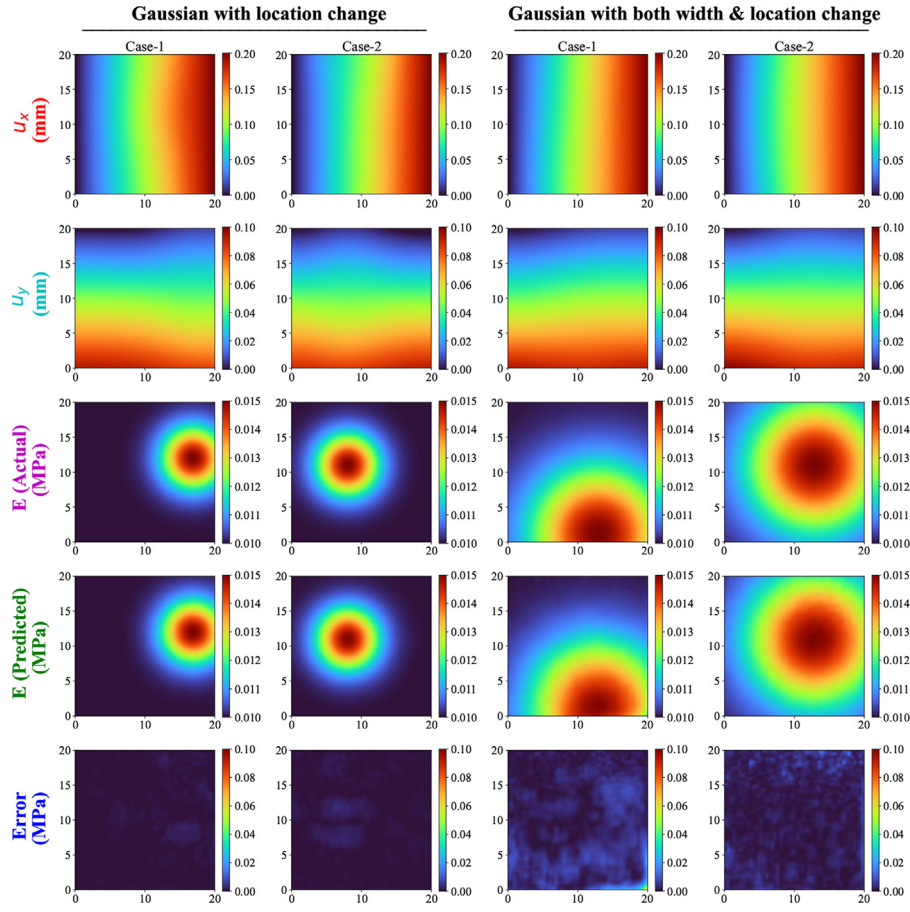
**Fig. 5. Displacement field to elasticity field learning process of the proposed WNO elastography.** (a) Training loss for all the cases of elasticity distribution considered in the present study with increasing heterogeneity in the distribution. (b) Evolution of elasticity prediction and relative error over training epochs for Gaussian type elasticity distribution. The error in the elasticity field mapping from input displacement data decreases, and the accuracy of the predictions increases with the number of epochs.

same as that of the proposed framework. The activation functions are chosen to provide the best performance of the CNN encoder-decoder network.

The model was trained firstly for the relatively simpler distributions of elasticity field (inclusion), followed by expanding it to complex distributions (sinc, 2D sinusoidal, and tumor dataset). The training loss at each epoch for the example problems is presented in Fig. 5a. It is observed that for the elastic fields having simpler distributions, the WNO converges more quickly than for the elasticity fields having complex distributions. However, within 500 epochs, even for the complex distributions, the WNO learns the operator responsible for mapping the displacement field to the elastic field. Along with the problem-specific training, the proposed framework is also trained using accumulated data from all the above examples. The accumulated data are sampled randomly for training and testing purposes. When learning the displacement to elasticity mapping for all the examples is carried out using a single network, the training errors are observed to be more, specifically  $\approx 3.0\%$ . The evolution of the elasticity prediction and associated error during the training phase are illustrated more in detail for the Gaussian inclusion type elasticity field in Fig. 5b. It can be seen

that as the number of epochs increases, the quality of prediction increases, and correspondingly the error between the predictions and the actual elastic field decreases.

The prediction results for the elasticity fields, along with the corresponding prediction errors for all the considered examples, are provided in Figs 6 and 7. The corresponding  $L^2$  relative error for each case is provided in Table 2. The results for the elastic field with Gaussian distribution are portrayed in Fig. 6. In the study with a change in the location of the Gaussian inclusion, it can be seen that the  $L^2$  relative error is less than 0.1%. The prediction error for the study with both location and width change of the Gaussian inclusion is observed to be less than 0.5%. The results for the complex elastic field, i.e., those with hard inclusion, sinc function, and the 2D Gaussian waves, are displayed in Fig. 7. In the prediction, the  $L^2$  prediction errors are observed to be around 1%, 1.6%, and 0.07%, respectively. Overall, in all the examples, the mean  $L^2$  relative error is observed to be  $< 2\%$  for most of the cases when the models are trained individually. However, when the testing is done using the model with accumulated training, the mean  $L^2$  relative error seems to be around 3%. In the last example, the elasticity field as a representative of benign and malignant tumors is taken.



**Fig. 6.** Learning the hidden elasticity field by utilizing the displacement field (in mm) in  $x$  and  $y$ -directions as the input obtained corresponding to the Gaussian type of distribution of Young's modulus (in MPa). The reference input and the corresponding predictions  $E(x, y)$  using the proposed WNO, along with the error between the two are also shown. (all axes are in mm).

**Table 2**

Prediction errors for the example problems.

| Examples                                                   | Mean $L^2$ relative testing error (%) |                     |
|------------------------------------------------------------|---------------------------------------|---------------------|
|                                                            | WNO                                   | CNN encoder-decoder |
| Gaussian with random location <sup>(3.1.2)</sup>           | 0.0954                                | 0.8193              |
| Gaussian with random location and width <sup>(3.1.2)</sup> | 0.3991                                | 2.2703              |
| Inclusion <sup>(3.1.1)</sup>                               | 1.0213                                | 5.6396              |
| Sinc <sup>(3.1.3)</sup>                                    | 1.6239                                | 7.4454              |
| 2D Sinusoidal <sup>(3.1.4)</sup>                           | 0.0686                                | 2.4467              |
| Tumor dataset <sup>(3.1.5)</sup>                           | 1.5460                                | 12.2831             |
| With accumulated training                                  | 2.9978                                | 14.7670             |

**Table 3**

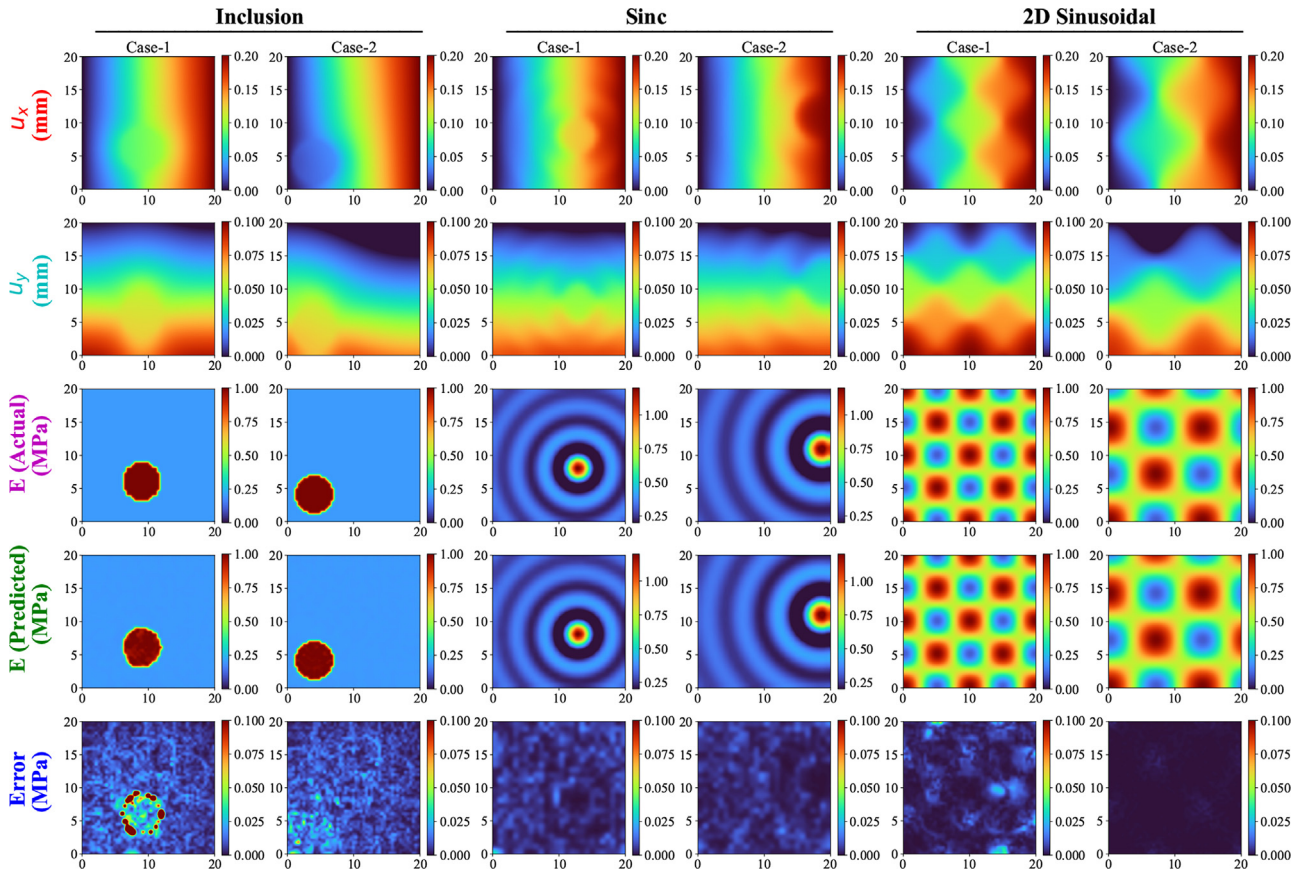
A comparison between the performance of the wavelet blocks with simple CNN blocks.

| Examples                                                   | Mean $L^2$ relative testing error (%) |                 |
|------------------------------------------------------------|---------------------------------------|-----------------|
|                                                            | WNO (with Wavelet Blocks)             | With CNN Blocks |
| Gaussian with random location <sup>(3.1.2)</sup>           | 0.0954                                | 12.6193         |
| Gaussian with random location and width <sup>(3.1.2)</sup> | 0.3991                                | 10.1253         |
| Inclusion <sup>(3.1.1)</sup>                               | 1.0213                                | 59.6197         |
| Sinc <sup>(3.1.3)</sup>                                    | 1.6239                                | 35.8047         |
| 2D Sinusoidal <sup>(3.1.4)</sup>                           | 0.0686                                | 29.4409         |
| Tumor dataset <sup>(3.1.5)</sup>                           | 1.5460                                | 76.9926         |
| With accumulated training                                  | 2.9978                                | 52.9527         |

This example, in particular, resembles the physical appearance of breast cancer tumors wherein the shape and elastic modulus of the tumor are one of the features used for its diagnosis. The benign tumors are homogeneous as compared to malignant tumors, which have heterogeneous elasticity distribution. Training the model on a mixture of both tumor cases is complicated because of the transition of elasticity values within the tumor itself. However, from the results in Fig. 8, it is evident that even for such a complex case, the  $L^2$  relative error is around 1.5%.

The prediction results of the CNN encoder-decoder network (similar to DLRE) are also presented in Table 2. We observe that for all the example problems including the accumulated training, the testing error for CNN encoder-decoder network is more than 4 times that for the proposed WNO-based elastography. We also

carried out an ablation study by replacing the four wavelet blocks with four CNN-Blocks (each block denotes a single CNN encoder-decoder network). The errors in the prediction results are compared with the proposed operator elastography in Table 3. It is observed that the prediction errors are very high for the latter case. This illustrates the efficacy of the wavelet blocks in the proposed architecture. In addition, we also found that optimization-based elastography method requires approximately 94 minutes to obtain the corresponding elasticity map for a single set of displacement fields. On the other hand, the proposed approach after training requires less than a second for prediction. This suggests that the proposed approach for using the wavelet neural operator



**Fig. 7.** Learning the hidden elasticity for hard inclusion (left), sinc (middle), and 2D sinusoidal (right) type of elasticity distributions (in MPa). The reference and the corresponding predictions  $E(x, y)$  using the proposed WNO, along with the error between the two are also shown. (all axes are in mm).

can be successfully used to directly map the elasticity field from the input displacement fields without sacrificing computational efficiency. The framework also circumvented the strain calculation part altogether, which involves differentiating the noisy displacement field data. The noise in the displacement data is amplified while calculating the strains, which ultimately affects the predictions by providing under or overestimation of the elasticity values. In the next section, the study on the soundness of the proposed scheme against observation noises is demonstrated.

### 3.3. Noise sensitivity

Signal corruption in the presence of environmental noise is a common phenomenon. To check the robustness of the framework against noise in the observations, a case study by adding the zero mean white Gaussian noise to the displacement field data is carried out in this section. The performance of the model has been tested against the noisy data by incorporating noise at different levels such as 1%, 2%, 5%, and 10%. The resulting signal-to-noise ratio ( $\text{SNR}_{\text{dB}}$ ) for the corresponding noise levels was found to be around 68, 62, 54, and 48 dB. The level of the noise indicates the percentage of the standard deviation of the actual displacement field data used for training. The earlier trained model with the clean displacement data has been used for testing with the noisy data. The error between the predictions and the actual elasticity fields is obtained using the  $L^2$  relative error in a similar manner as before. The test results are demonstrated in Fig. 9 and Table 4. The results are also compared with the results of the previously defined CNN encoder-decoder network, which are tabulated in Table 4.

**Table 4**

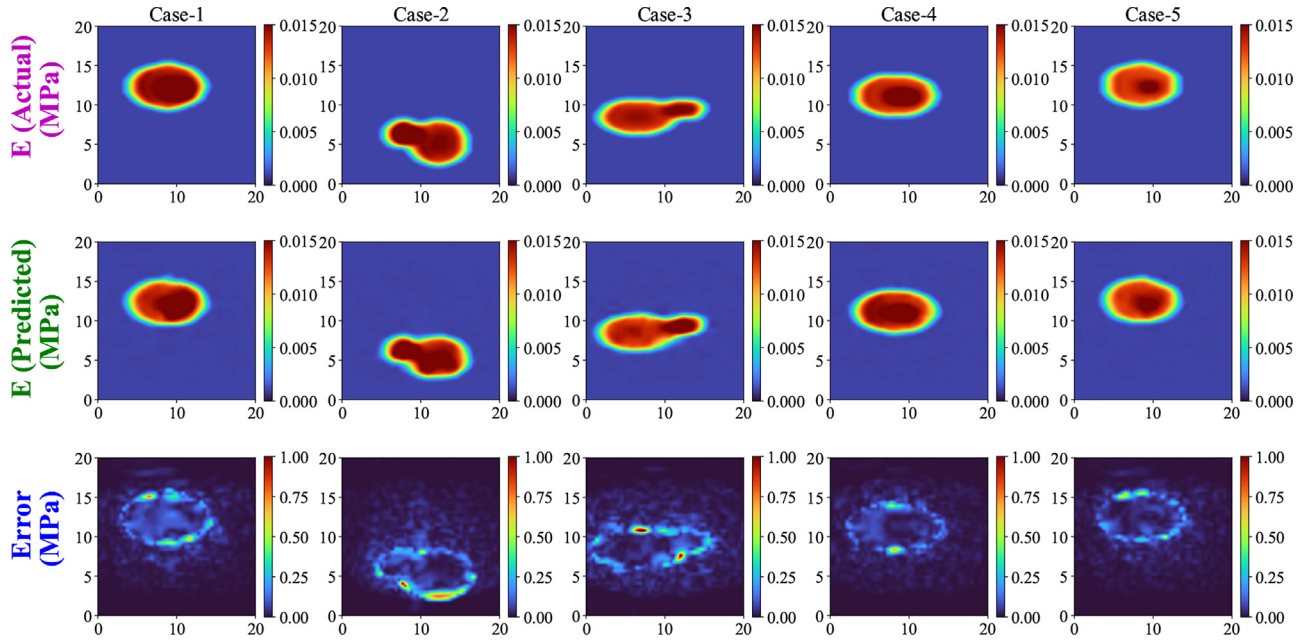
Error in the predictions for example problems.

| Noise<br>(in % of the stdev.) | SNR (dB) | $L^2$ relative error (%) |                     |
|-------------------------------|----------|--------------------------|---------------------|
|                               |          | WNO                      | CNN encoder-decoder |
| 1                             | 68       | 0.4199                   | 2.1210              |
| 2                             | 62       | 0.4212                   | 2.1249              |
| 5                             | 54       | 0.5166                   | 2.1735              |
| 10                            | 48       | 0.5628                   | 2.2251              |

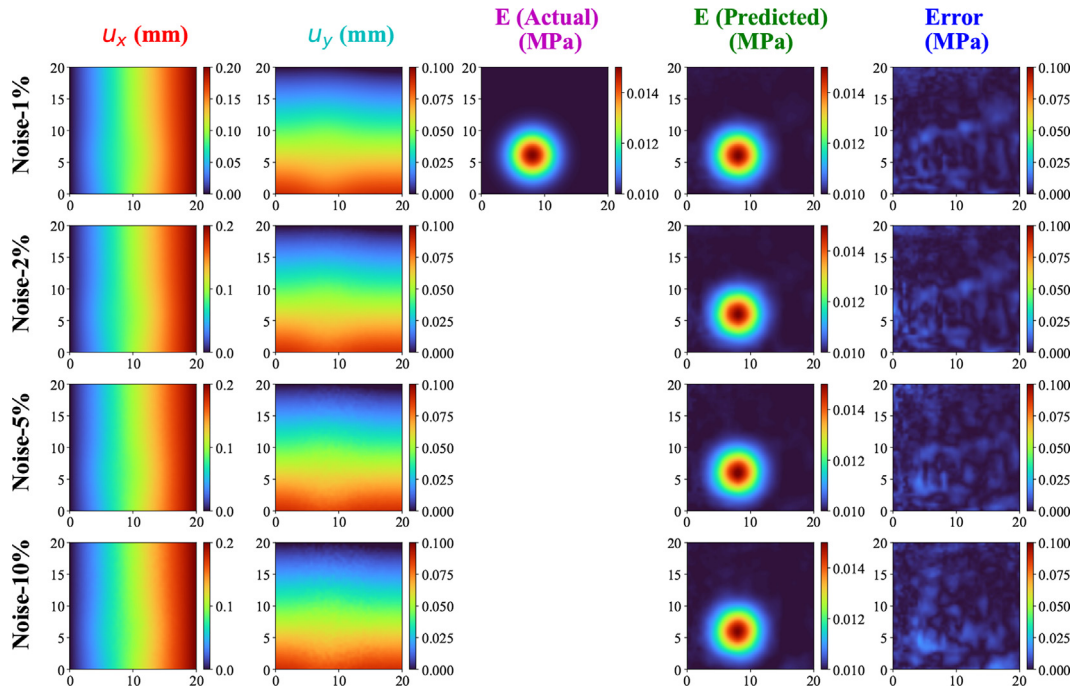
In Fig. 9, the noisy displacement field based on different noise levels and the actual and predicted elasticity fields are provided along with the prediction error. From the results, it is observed that with increases in the noise level, the prediction errors for both the proposed WNO-based elastography and simple CNN encoder-decoder based elastography (one inspired by DLRE) increase. Although the increase in mean  $L^2$  relative error is minimal and the overall fluctuations are quite stable, the proposed WNO-based elastography outperforms the CNN encoder-decoder. This can be attributed to the fact that the parameterization of the neural operator is done in the last level of the multi-resolution wavelet space. In the multi-resolution space, the noises get filtered out in the primary levels allowing the model to learn the noise-free features of the input data. This clearly evidences the ability of the proposed framework to handle the presence of noise in the observations.

### 3.4. Experimental verification

For the demonstration of the proposed framework to real-world data, an example of a deformed elastography phantom (CIRS elas-



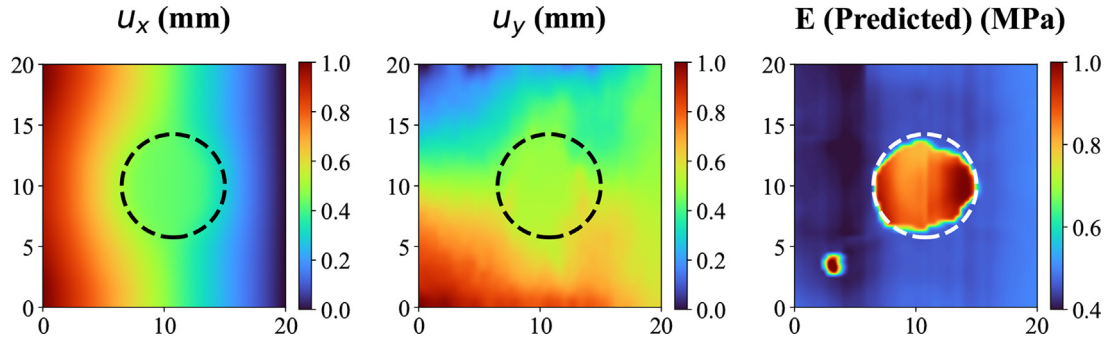
**Fig. 8.** Application of WNO for the test cases of benign and malignant tumor classes detected by the WNO. The reference input (top row) and the corresponding predictions (middle row) along with the error (bottom row) are shown. (all axes are in mm).



**Fig. 9.** Effect of noise on the prediction of WNO at different noise levels. The reference input and the corresponding predictions  $E(x, y)$  (in MPa) using the proposed WNO, along with the error between the two is shown. (all axes are in mm).

tography phantom, Norfolk VA) embedded with a spherical stiff inclusion is considered. The embedded inclusion is approximately 1.8 times stiffer than the soft background. The deformation field was acquired using 2D Ultrasound (US) compression quasi-static US elastography technique [47,48]. In this technique, two B-mode US images of region of interest are taken, one before and another one after the application of compressive force. These B-mode images of a compressed and uncompressed phantom are then used to reconstruct the displacement field data in the axial and lateral directions using methods such as window-based, iterative or deep-learning based [9,49]. In the present case, the displacement field was recon-

structed using a variant of window-based framework named tTotal Variation rEgularization and WInDow-based time-delay estimation (OVERWIND) proposed by Mirzaei et al., [49]. OVERWIND takes the advantages of window-based and optimization based methods with total variation regularization by combining both into a single objective function to yield a robust framework for displacement estimation. Due to the high resolution of US images in the axial direction, the measured displacement field was more accurate in the axial direction than in the lateral direction (refer Fig. 10) [50]. In strain-based elastography, the stiff inclusion experiences less strain for uniformly applied stress than the soft background. The strain



**Fig. 10.** Displacement field data acquired using US-based elastography in the (left) axial and (middle) lateral directions with the corresponding prediction of elasticity field (right) highlighting the stiff inclusion. The ratio between the average stiffness of the inclusion and soft homogeneous background was found to be two.

maps thus generated using the 2D displacement field are overlaid on B-mode images to aid the assessment of lesions qualitatively. Since the local stress values are unknown, only a relative but essential stiffness difference can be obtained.

In the present work, the noisy displacement field is utilized directly to test the model trained using the stiff inclusion data described in Section 3.1. We again emphasize that no separate training for the experimental data was utilized and earlier trained model was used for the prediction. The measured displacement field was interpolated to match the spatial resolution of the training dataset before testing. In addition, no noise filtering was applied to the acquired data, as the primary levels in the WNO were sufficient to eliminate the noise. Based on the resulting displacement field data in axial and lateral directions, the elasticity prediction highlights the location of stiff inclusion (refer Fig. 10). The inclusion was found to be two times stiffer than the soft background in an average sense. However, the complete shape of the inclusion is not recovered completely because of inaccuracies in the data measured in the lateral direction ( $u_y$ ) [50]. The prediction can be improved by training the model with data specific to the US elastography. Although, due to the paucity of experimentally available data, simulation data as a representative of the clinical data can also be utilized [51].

#### 4. Discussions and conclusion

This paper proposes a novel method of elastography using a deep neural operator that directly maps the elastic properties from the displacement field. The proposed method uses the concept of recently proposed neural operators. In particular, we exploit the superiority of the wavelet neural operator to map the displacement fields to elastic fields. The neural operators act as functions that map the input function space to output function spaces. We utilized this fact to construct a mapping between the displacement and elastic fields. The proposed elastography is highly robust and advantageous to iterative and finite element-based methods. Once trained, the proposed method can identify the exact elastic properties within a fraction of a second. The proposed method is also found to be superior to CNN-based elastography as the proposed method learns the underlying operator rather than a functional mapping between displacement to elasticity fields. A unique approach is taken to learn the mapping, where the neural network is parameterized in wavelet space rather than direct parameterization in physical coordinates. In the wavelet space, the frequency information is localized spatially, allowing efficient and accurate learning of the exact pattern of the input. Further, a multi-resolution treatment is given to the input displacement field in which the parameterization is performed on the last level of the multi-resolution wavelet decomposition. This greatly helps in the

isolation of the key information of the features from the observation noise.

In our experiments, we have considered several numerical case studies involving various types of elasticity field distributions. A problem for diagnosing benign and malignant type tumors from the healthy tissues based on their shape and elasticity distribution is also included. In each example, we have demonstrated that our proposed approach can reproduce the elasticity field from the input displacement data with high accuracy. Once trained, the proposed method also generalizes to new inputs. The same has been demonstrated using experimental data wherein a pre-trained model is used to predict the elasticity field of the experimentally measured displacement field data. The experimental result demonstrates that the proposed method can produce the target elasticity field with sufficient accuracy. In all the cases, the proposed framework is trained only for 500 epochs using 500 training data sets. This indicates that the proposed method for elastography is very data efficient. The proposed elastography is expected to be convenient wherever the exact physics for the corresponding stiffness-displacement relationship is not known. In the present work, the Poisson's ratio was assumed to be 0.3 for all the simulations. However, the behaviour of soft tissues is nearly incompressible in nature with Poisson's ratio approaching a value close to 0.5. The high bulk modulus resulting from such high Poisson's ratio is tackled numerically using displacement-pressure-based mixed formulation. Consequently, only displacement field data can be utilized, neglecting the pressure field information which is unavailable in most elastography applications. As the current framework learns the mapping between displacement field data and respective elasticity distribution, we believe that the framework can be re-trained utilizing displacement field data only.

Due to the advancement in medical imaging techniques such as magnetic resonance imaging (MRI) and ultrasonography, it is now possible to obtain full-field deformation maps and corresponding elastograms of various parts and/or organs of the human body. These techniques come under the umbrella of static and dynamic elastography; wherein time-resolved displacement field data are recorded as the shear wave passes through the tissue. In MR Elastography, the amplitude and phase data corresponding to the fundamental frequency (or first harmonic) is extracted while the rest of the data at higher harmonics is discarded. This results in loss of information as studies have demonstrated the contribution of data captured at higher harmonics [52]. Alternatively, in shear wave elastography, time-resolved shear wave images are directly used for the reconstruction of elasticity [53]. In such cases, the proposed neural operator-based elastography can be alternatively applied to deal with time-resolved motion data directly. This approach would correctly interpret the stiffness distribution of tissues without loss of information. In future works, we aim to extend the present work

for dynamic elastography cases by utilizing the time-resolved displacement field data. Additionally, the proposed framework can be extended by integrating the known physics into the proposed neural operator for elastography [54,55]. This will increase the generalizability of the proposed framework for elasticity fields with unseen and complex distributions. The learned data from pre-trained models can also be employed for transfer learning, which reduces the effective training time for models with random initialization [56]. Classification of medical images using the proposed framework can be another potential extension. Further, the proposed approach shall aid in the real-time diagnosis of medical images.

### Code availability

The source codes and data to reproduce the results reported in this study can be downloaded from: <https://www.github.com/csccm-iitd/WNO-elastography>.

### Declaration of competing interest

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

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